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Magnetostratigraphy of the Sherwood Sandstone Group (Lower and Middle Triassic), south Devon, UK: detailed correlation of the marine and non-marine Anisian

Mark W. Hounslow [a,*], Gregg McIntosh [b]

[a] *Centre for Environmental Magnetism and Palaeomagnetism, Geography Department, Lancaster Environment Centre, Lancaster University, Lancaster LA1 4YW, UK*

[b] *Departamento de Geofisica y Meteorologia, Facultad de CC Fisicas, Universidad Complutense de Madrid, Avenida Complutense, 28040 Madrid, Spain*

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Abstract

The Sherwood Sandstone Group in south Devon consists of the Budleigh Salterton Pebble Beds and the Otter Sandstone Formation. The Otter Sandstone Formation comprises four lithostratigraphic sub-units, from the base labelled A–D, and contains tetrapods corresponding to the Perovkan land vertebrate faunachron, and is consequently a key European non-marine Anisian succession. A total of 181 palaeomagnetic specimens from 92 horizons were measured to determine the palaeomagnetic signal. The magnetic properties and characteristic remanence is predominantly carried by haematite, most of which probably formed early in the depositional and diagenetic history. The lower part of the Sherwood Sandstone Group (Budleigh Salterton Pebble Beds and Otter Sandstone Formation, unit A, lower part of unit B) has predominantly normal polarity (magnetozones BS1, BS2 and BS3n) with three thin reversed polarity magnetozones. The lower part of unit B to the lower part of unit C of the Otter Sandstone Formation is predominantly reverse polarity (magnetozones BS3r, BS4 and BS5), but includes three short normal polarity intervals. The upper ca. 50 m of unit C shows a mix of normal and reverse polarity (BS6, BS7) with normal dominating. The ca. 15-m-thick unit D is entirely of normal polarity (BS8n). The magnetostratigraphy allows unambiguous comparison with conodont and ammonite calibrated magnetostratigraphies from elsewhere in Europe. These show that the BS2 and BS3n magnetozones of the Otter Sandstone Formation are lower Aegean to middle Pelsonian (lower to middle Anisian). The BS3r to BS5r magnetozone interval, is middle Pelsonian to lower Illyrian (middle to upper Anisian), and the BS6 to BS8n magnetozone interval is middle Illyrian to lower Fassanian. The top of unit D is correlated to an interval in the middle of the Nevadites Zone (close to the Anisian–Ladinian boundary). Consequently, the Otter Sandstone Formation correlates with the Muschelkalk in the Polish Basin, but is equivalent to only part of the Germanic Basin Muschelkalk. The continuous magnetostratigraphy from the Sherwood Sandstone Group, provides additional details of the polarity pattern through the Pelsonian to Illyrian interval, which appears to be lacking in discontinuous marine sections. A synthesis of adjacent European sedimentary basins in combination with the magnetostratigraphy, suggests that the Budleigh Salterton Pebble Beds are probably Late Spathian to lower Aegean.
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* Corresponding author. Tel.: +44-(0)-1524-594588. *E-mail address:* m.hounslow@lancaster.ac.uk (M.W. Hounslow).

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1. Introduction

The onshore Triassic in Britain differs from that in the southern North Sea, Germany and other parts of northern Europe, in not containing a tripartite lithostratigraphic subdivision into the classic Buntsandstein, Muschelkalk and Keuper. Instead, the British Triassic is divided largely into the Sherwood Sandstone Group and the overlying Mercia Mudstone Group which together attain a maximum onshore thickness of ca. 3.5 km in the Cheshire Basin (Warrington et al., 1980; Evans et al., 1993; Fig. 1A). The relationships between the predominantly non-marine British Triassic, the classic German lithostratigraphy and the Tethyan biostratigraphy is understood in outline, but the paucity of fossil evidence, hinders detailed correlation. In Britain, Triassic biostratigraphy is either based on sparse vertebrate data, or more commonly on miospores. The magnetostratigraphy of the Sherwood Sandstone Group in Devon is important, in that it provides a means for establishing detailed links between the conodont-calibrated magnetostratigraphy from the Tethys (Muttoni et al., 1997a; Muttoni et al., 2000) and the tetrapod biostratigraphy of continental successions. This work contributes to an improved understanding of western European inter-basin correlations during the Anisian.

2. Stratigraphy

2.1. Lithostratigraphy

In the Wessex Basin, the Sherwood Sandstone Group is divided into a lower conglomeratic unit, the Budleigh Salterton Pebble Beds and an overlying sandstone-dominated unit, the Otter Sandstone Formation (Fig. 2). These units outcrop in Devon and are well-exposed on the south Devon coast. The Budleigh Salterton Pebble Beds are composed of ortho-conglomerates and a small proportion of medium-grained sandstones (Figs. 3A and 4). These are the deposits of a braidplain or wet alluvial fan, which largely drained the Armorican land mass located in western France and the southern English Channel, and transported exotic clastics northwards into the Wessex Basin (Holloway et al., 1989; Smith, 1990; Smith and Edwards, 1991). The boundary between the Budleigh Salterton Pebble Beds and the underlying Aylesbeare Mudstone Group is generally considered to be an unconformity (Holloway et al., 1989), although this is most clearly apparent from seismic surveys and well data further east in the Wessex Basin (Butler, 1998; Fig. 1A). In the deeper parts of the Wessex Basin, seismic studies suggest that the base of the Otter Sandstone Formation marks a change in the style of basin tectonics, and that additional units are preserved locally between the Budleigh Salterton Pebble Beds and Otter Sandstone Formation. Consequently, the base of the Otter Sandstone Formation has been equated with the Hardegsen disconformity of the Germanic Basin (Holloway et al., 1989; Butler, 1998).

The Otter Sandstone Formation largely consists of stacked fluvial channel sandstones, composed of medium- to fine-grained, occasionally muddy sandstone (Fig. 3C,D). The basal beds of the channel sandstones have channel lags of mudstone, exotic gravel clasts (quartz and slate fragments) or most commonly, locally reworked calcrete (Fig. 3B). Notable is the relative absence of overbank mudstone deposits, from which the calcretes and mudstone clasts are probably derived.

In coastal outcrops, the Otter Sandstone Formation is divisible into four units. Unit A, a heterogeneous basal unit (Fig. 4), consists of a lower 7 m of aeolian sandstone resting on a reg palaeosol with ventifacts (Fig. 3A; Smith, 1990; Wright et al., 1991), overlain by fine- to medium-grained, fluvial channel sandstones with no calcrete (Fig. 4). The ventifact horizon appears to be a consistent feature of the top of the Budleigh Salterton Pebble Beds in exposures further north (Smith and Edwards, 1991), and the reg palaeosol has

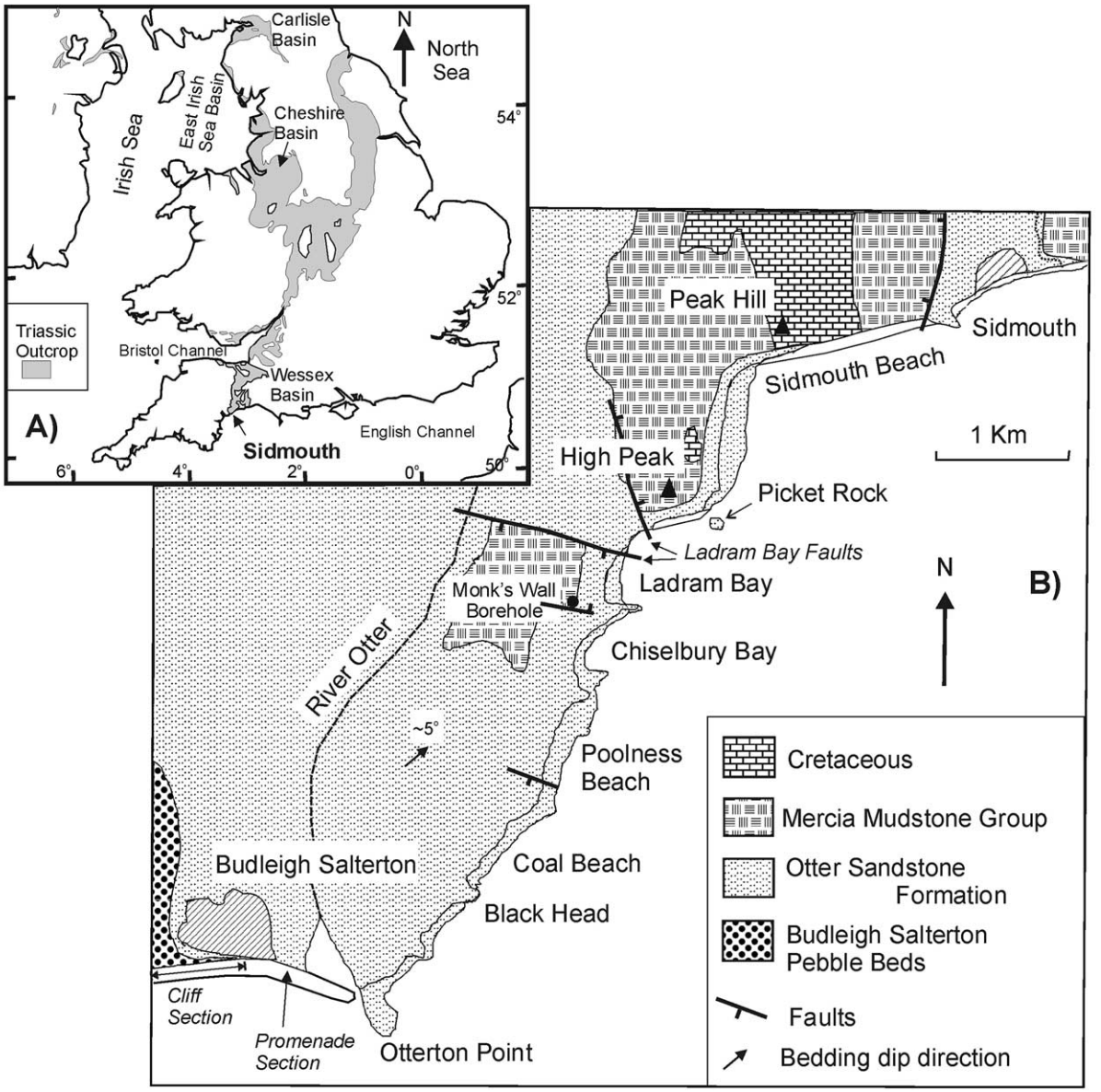


Fig. 1. (A) Study location. (B) Outline geology and section locations between Budleigh Salterton and Sidmouth (modified from Benton, 1997).

been interpreted as evidence of a substantial hiatus (Wright et al., 1991).

Unit B is characterised by abundant reworked calcrete in the channel sandstones (Figs. 5 and 6), and vertical calcrete structures (rhizoconcretions) formed around the taproots of phreatophytic plants (Purvis and Wright, 1991; Fig. 3B,C). Horizontal sheets of calcrete are also present and were formed at Triassic water table surfaces in the floodplain sediments (Purvis and Wright, 1991).

Unit C has sparse development of vertical and horizontal calcretes, but mudstone and sandstone filled channels are more common, in comparison to the underlying unit B (Figs. 3D, 6 and 7). Unit

	Formation (*=biostratigraphic age control)
Mercia Mudstone Group	Blue Anchor Fm (*Rhaetian) {24 m} Twyning Mudstone Fm {282 m} Arden Sandstone Fm (*Carnian) {32 m} Elderfield Mudstone Fm {165 m}
Sherwood Sandstone Group	Otter Sandstone Fm (*Anisian) {210 m} Budleigh Salterton Pebble Beds {32 m}
Aylesbeare Mudstone Group	Littleham Mudstone Fm {275 m} Exmouth Sandstone and Mudstone Fm {255 m}
Exeter Group+	Dawlish Sandstone {100-120 m} Monkerton Fm {0-40 m} Heavitree Breccia {135-300 m} Alphington Breccia {160-240 m} Whipton Fm (*Upper Permian) {20-60 m}

Fig. 2. Lithostratigraphy of the Permian–Triassic in the western part of the Wessex basin. Thickness in metres (from Hamblin et al., 1992; Edwards and Scrivener, 1999). +In the Exeter area.

D is a mudstone-rich unit, devoid of calcrete conglomerates and transitional in character to the Mercia Mudstone Group (Fig. 7). The sandstones in the lower part of unit D have well developed, N–S oriented sandstone and mudstone filled channels 2–30 m wide and up to 4 m deep. The unit C–unit D boundary may be a minor disconformity, since the base of unit D cuts across sandstone bodies in unit C. The Mercia Mudstone Group represents the deposits of a predominantly lacustrine playa system, dominated by red, occasionally gypsiferous, mudstones (Hamblin et al., 1992; Talbot et al., 1994).

2.2. Biostratigraphy

The Otter Sandstone Formation is important because it contains tetrapod faunas, that have been used for both regional and global correlations (Benton, 1997; Lucas, 1998). The fauna is dominated by reptilian, rhynchosaur remains, but includes temnospondyl amphibians and fish (Milner et al., 1990; Spenser and Storrs, 2002). Elements of this association indicate age correlations with the Donguz Formation in the Russian Urals, the Upper Buntsandstein (Röt Formation in Germany), the Holbrook Member (Moenkopi Formation, USA), and the Ermaying Formation in the Ordos Basin (China), and thus indicate an Anisian age (Benton, 1997; Lucas, 1998; Spenser and Storrs, 2002). The Perovkan land vertebrate faunachron defined by Lucas (1998), also has elements in common with the Otter Sandstone Formation fauna. However, the Otter Sandstone Formation assemblage is unusual in two ways: (1) it lacks mammal-like reptiles (synapsids) which are common in North American and Gondwana assemblages of Middle Triassic age, and (2) like other British Triassic faunas, it is dominated by rhynchosaurs (Benton and Spencer, 1995), which are most abundant in Gondwana Triassic units, but are rare or absent at other European or North American Triassic sites. The Otter Sandstone Formation therefore provides a critical palaeontological link between these widely separated geographic locations.

Tetrapod remains in the Otter Sandstone Formation most commonly comprise disarticulated skull elements found in the calcrete-rich channel lag conglomerates (Milner et al., 1990; Benton, 1997). Most of the remains have been found in unit C, particularly towards the top, although some have come from the lower part of unit B at Otterton Point (Fig. 1B; Benton, 1997; Spenser and Storrs, 2002).

The youngest age limit of the Otter Sandstone Formation is constrained by a unit of colour-banded mudstones within the Mercia Mudstone

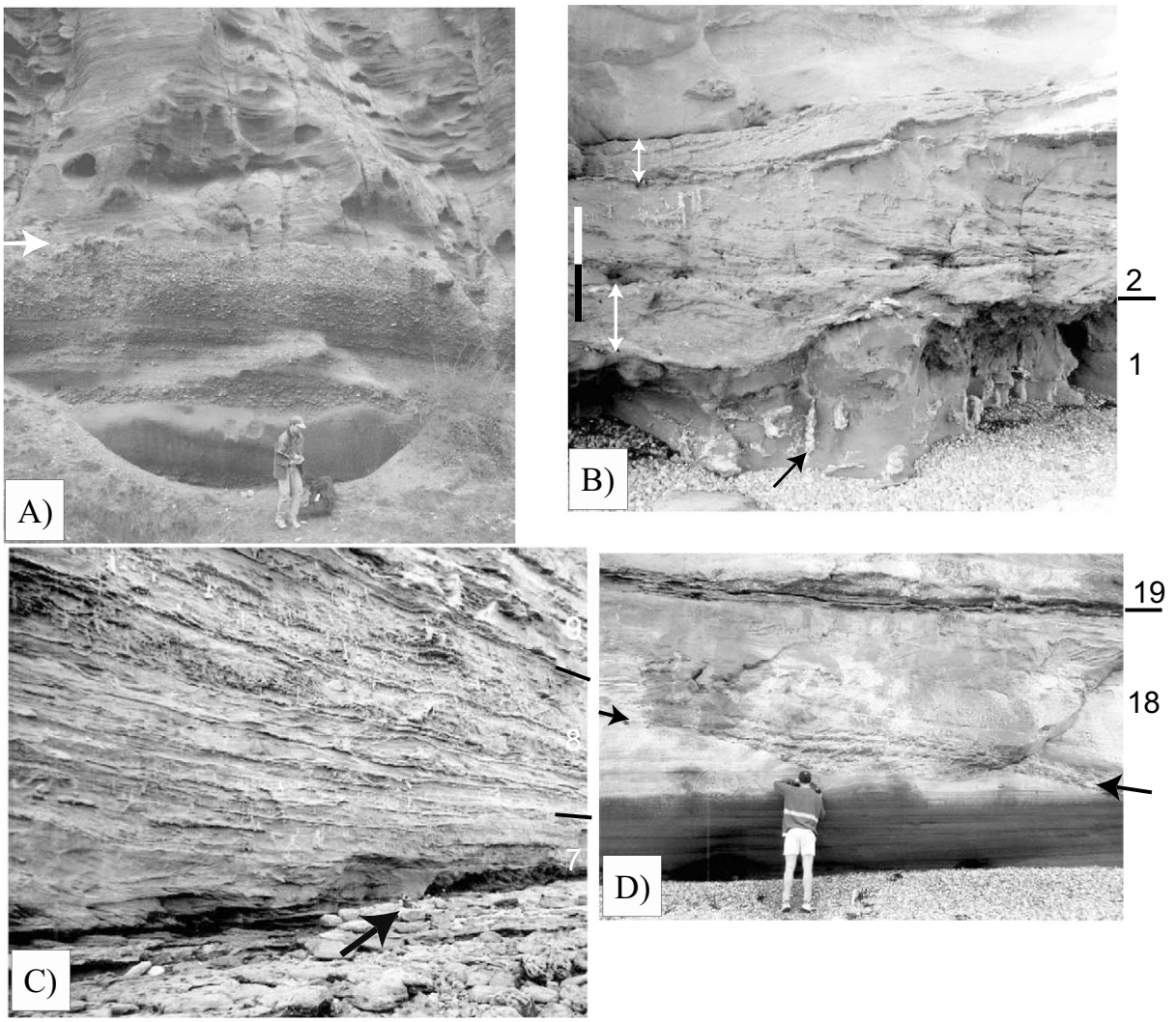
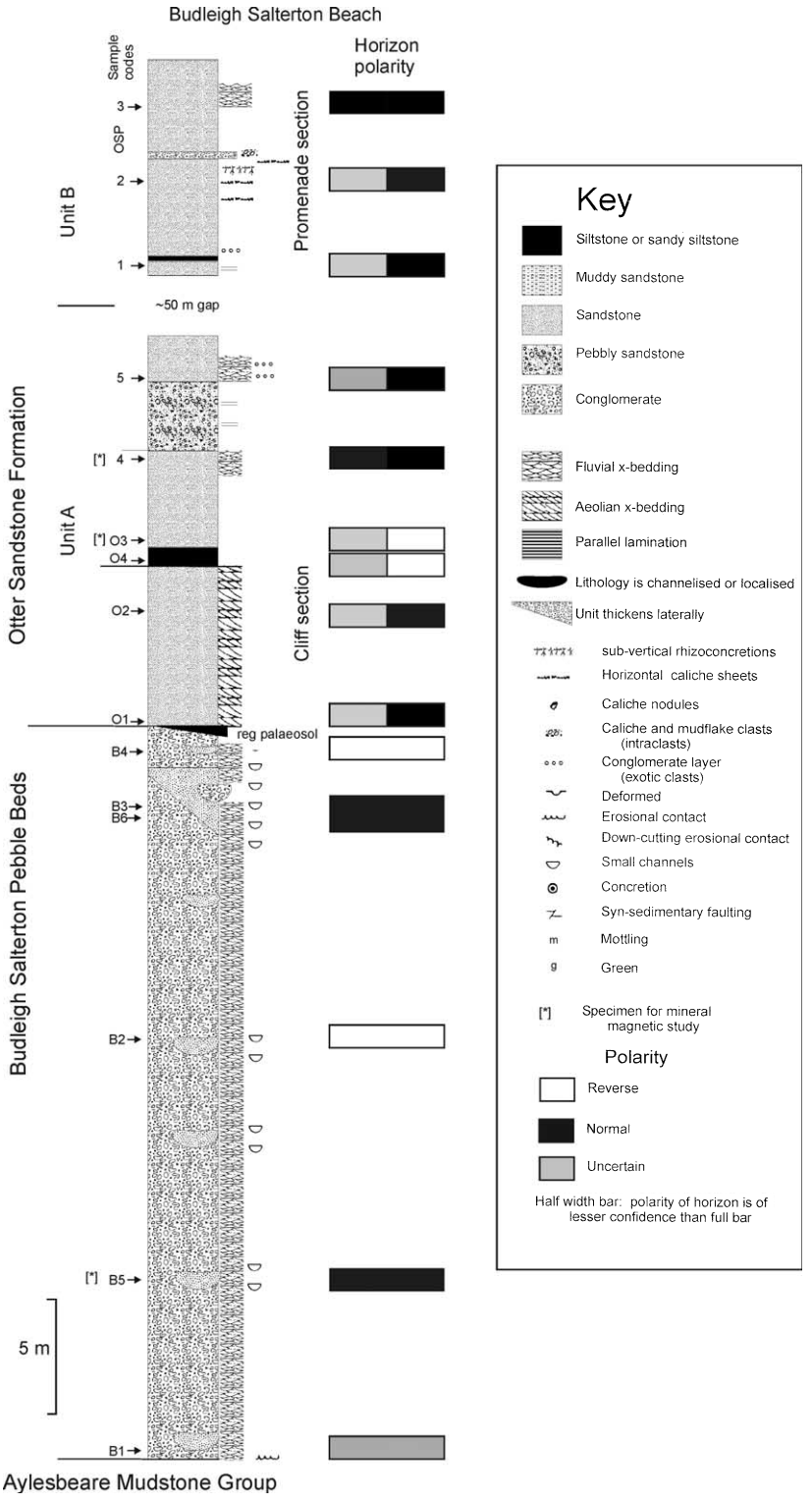


Fig. 3. Typical sediment architecture of the Sherwood Sandstone Group. In the case of (B), (C) and (D) the photo-layer boundaries are marked. (A) Uppermost part of the Budleigh Salterton Pebble Beds at its junction (marked by arrow) with the aeolian cross bedded sandstone, in the lowest part (unit A) of the Otter Sandstone Formation. (B) Lower part of unit B (Otterton Point; Fig. 1B), showing the boundary between photo-layers 1 and 2. The base of layer 2 is marked locally by two conglomerate beds (double-headed arrows), mostly composed of re-deposited calcrete. Rhizoconcretions (arrowed) are common in layer 1. Vertical scale sub-divisions 0.5 m. (C) Bounding surfaces between photo-layers 7, 8 and 9, with extensive development of rhizoconcretions in layer 8. Arrow indicates two persons for scale. (D) Down-cutting relationship (arrowed) of channel sandstone within the top of photo-layer 18, Ladram Bay.

Group (probably equivalent with the Arden Sandstone Formation; Fig. 2) which contains a miospore assemblage assigned to the Carnian (Warrington, 1997). Traditionally, the Budleigh Salterton Pebble Beds have been equated with similar units in the lower part of the Sherwood Sandstone Group in the English Midlands and Irish Sea Basin, because of similar exotic clast content (Warrington et al., 1980; Holloway et al., 1989; Meadows and Beach, 1993). Biostratigraphic constraints on the age of the Budleigh Salterton Pebble Beds are provided only by the


Budleigh Salterton Beach
Sample codes
Horizon polarity
Promenade section
Cliff section
Unit B
Unit A
OSP
3
2
1
~50 m gap
5
[*] 4
[*] O3
O4
O2
O1
reg palaeosol
B4
B3
B6
B2
[*] B5
B1
5 m
Otter Sandstone Formation
Budleigh Salterton Pebble Beds
Aylesbeare Mudstone Group
Key
Siltstone or sandy siltstone
Muddy sandstone
Sandstone
Pebbly sandstone
Conglomerate
Fluvial x-bedding
Aeolian x-bedding
Parallel lamination
Lithology is channelised or localised
Unit thickens laterally
sub-vertical rhizoconcretions
Horizontal caliche sheets
Caliche nodules
Caliche and mudflake clasts (intraclasts)
Conglomerate layer (exotic clasts)
Deformed
Erosional contact
Down-cutting erosional contact
Small channels
Concretion
Syn-sedimentary faulting
m Mottling
g Green
[*] Specimen for mineral magnetic study
Polarity
Reverse
Normal
Uncertain
Half width bar: polarity of horizon is of lesser confidence than full bar

Anisian age tetrapods from the Otter Sandstone Formation and Upper Permian miospores from the Exeter Group (Fig. 2; Warrington and Scrivener, 1990; Edwards et al., 1997; Edwards and Scrivener, 1999). Tetrapod footprints (*Chelichnus bucklandis*) in the Dawlish Sandstone (Edwards and Scrivener, 1999) have equivalents in the uppermost Rotliegend of Germany and are therefore also compatible with an Upper Permian age for the beds lower in the Exeter Group (Edwards et al., 1997).

3. Methodology

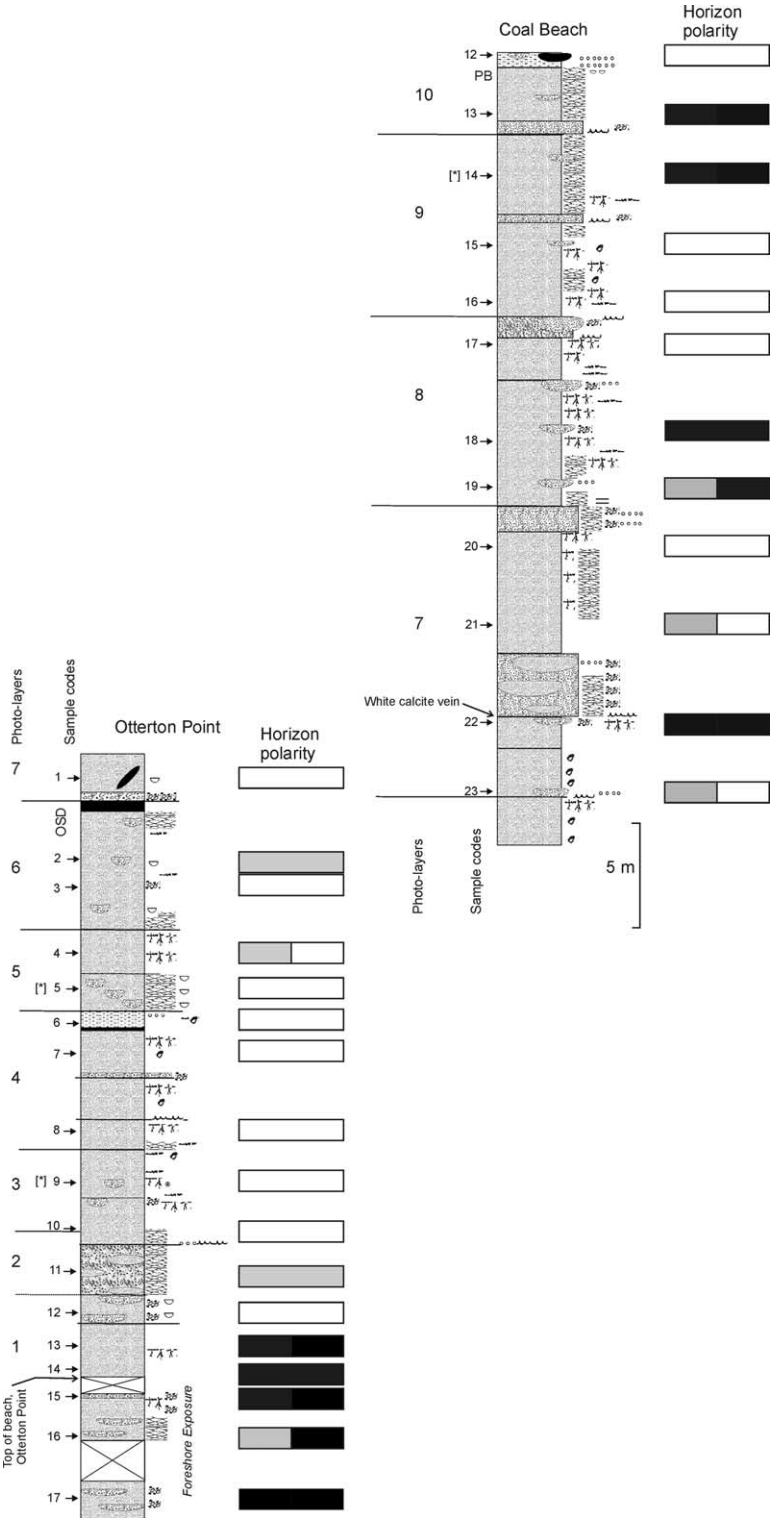
3.1. Sampling

Samples were collected from 92 horizons through the Sherwood Sandstone Group, with the sampling horizons located onto a detailed log of the sections (Figs. 4–7). Two sections were sampled west of the River Otter (Figs. 1B and 4), the first in the cliffs directly below the town of Budleigh Salterton (cliff section), and the second, eastwards towards the main beach car park (promenade section). The estimated stratigraphic gap between these sections is ca. 50 m, based on a bedding dip of 5°. The Budleigh Salterton Pebble Beds and its transition into the Otter Sandstone Formation is exposed in the cliff section (Fig. 3A).

The Otter Sandstone Formation east of the River Otter has continuous exposure with a generally consistent dip of 5–10° to the NE. However, access to much of the exposure requires a boat during low tide. Detailed lithological logs were constructed by the authors from individual sub-sections, onto which the sampling horizons were located. Intersection correlation was made possible by a series of overlapping photographs of the cliff exposures between Otterton Point and Ladram Bay (Fig. 1B). In these photos, 21 layers could be defined on the basis of conformity of prominent planar surfaces (Figs. 3, 5–7). These surfaces generally bound 1–4-m-thick sandstone sheets, that have greater than ca. 400 m lateral continuity, composed of amalgamated channel deposits (Fig. 3C). The most prominent bounding surfaces were used for defining the 'photo-layer' boundaries. The photographs and layer interpretations were provided by A. Newell and R.A. Edwards of the British Geological Survey. The boundary of unit B and unit C occurs within photo-layer 12 (Fig. 6). Sandstones from layers 13 to 21 (i.e. unit C) generally lack well developed vertical calcrete rootlet systems (exceptions are layers 17 and 19) but have common siltstone channel fills near the top of the fining-up cycles (Figs. 6 and 7). East of the faults just NE of Ladram Bay (Fig. 1B), the photo-layers are no longer discernible, because of down-wash from the overlying mudstones. The top of the uppermost photo-layer 21 (Fig. 7) is interpreted to mark the base of unit D, which forms a distinct lithological boundary in the cliffs NE of Ladram Bay. The section from Picket Rock eastwards (Figs. 1B and 7) shows three major bounding surfaces, which are interpreted as the photo-layers 19, 20, and 21. Stratigraphic and sampling overlap occurs across photo-layer 7 (Fig. 5), layers 17 and 18 (Figs. 6 and 7) as well as 18 and 19 (Fig. 7).

Lithological correlation is not possible between the sections east and west of the River Otter estuary and it is probable that N–S oriented faults are present. Consequently, the thickness of the Otter Sandstone Formation cannot be fully evaluated at outcrop. A borehole at Monk's Wall (northwest of Chiselbury Bay; Fig. 1B), indicates that the Otter Sandstone Formation is 210 m thick (A.J. Newell, pers. commun.). The cumulative section thickness of 210 m for the Otter Sandstone Formation has been measured, indicating that west of the River Otter the lower part of unit B is probably repeated by faults with a net down throw to the west. The promenade section appears to be equivalent to the lower part of unit B.

Fig. 4. Sedimentological log of the lowest part of the Sherwood Sandstone Group at Budleigh Salterton. The log of the Budleigh Salterton Pebble Beds is a simplified version, redrawn from Smith (1990). On the left hand side of the log are the magnetostratigraphic sampling horizons (with sample codes indicated), and on the right the polarity of that horizon.


Coal Beach
Horizon polarity
PB
White calcite vein
Photo-layers
Sample codes
5 m
Otterton Point
OSD
Top of beach, Otterton Point
Foreshore Exposure

Conventional palaeomagnetic drilling methods were unsuccessful in the Sherwood Sandstone Group because of the friable nature of the sandstones. Hand-samples were cut from suitable exposures, using hammer and chisels, mostly at the base of the cliff, or from exposures on the wave-cut platform. A suitable flat surface on each specimen was oriented using a magnetic compass and a specially designed orientation staff. Specimens were re-oriented in dental plaster and cubic palaeomagnetic specimens were cut with a diamond saw. About one third of the samples were cut with no saw coolant, because they were not sufficiently strong to withstand wetting, and a similar number were cut wet. The remainder were too fragile to cut, so they were impregnated with a sodium silicate solution, which was later hardened by heating at 40°C. In this hardened state palaeomagnetic specimens could be cut from the samples using a diamond saw.

3.2. Magnetic measurements

Low field magnetic susceptibility (k) was measured using either a Bartington MS2B sensor or an AGICO KS3 Kappabridge. Measurements of Natural Remanent Magnetisation (NRM) were made using a CCL GM400 3-axis cryogenic magnetometer (noise level ca. 0.002 mA/m). Multiple specimens from the hand-sample at each horizon were subjected to stepwise thermal demagnetisation, using a Magnetic Measurements Ltd. thermal demagnetiser in 25–100°C steps up to 680°C, with steps more closely spaced above 400°C. The susceptibility measurements were performed after each heating step, to monitor mineralogical changes

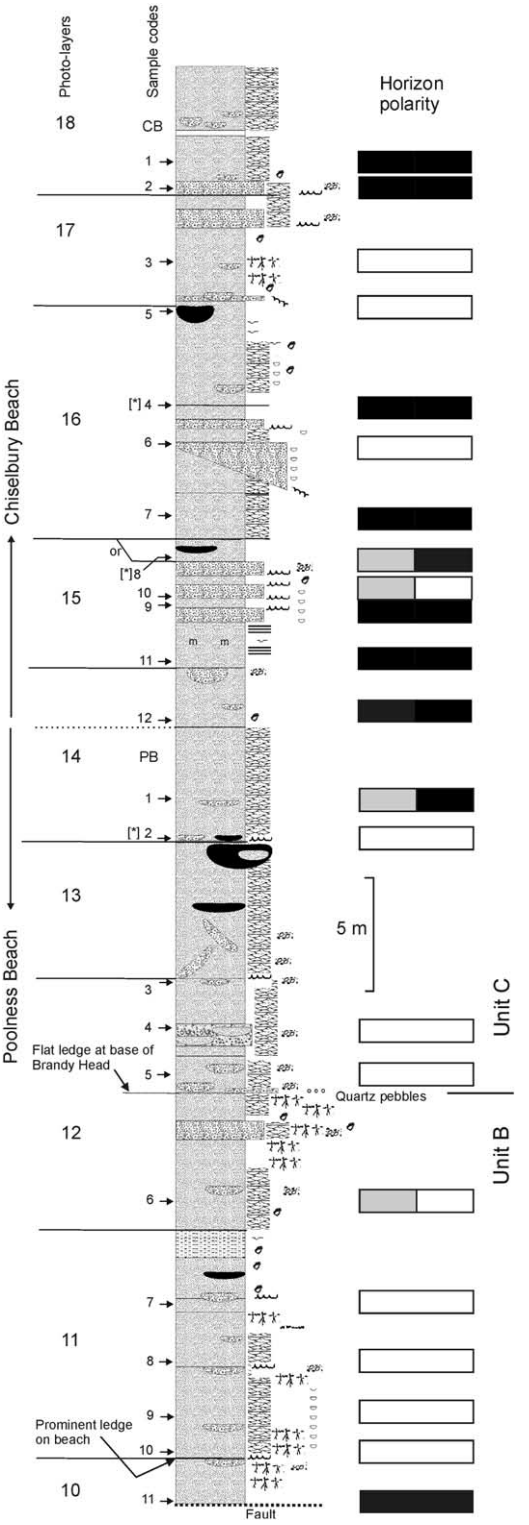
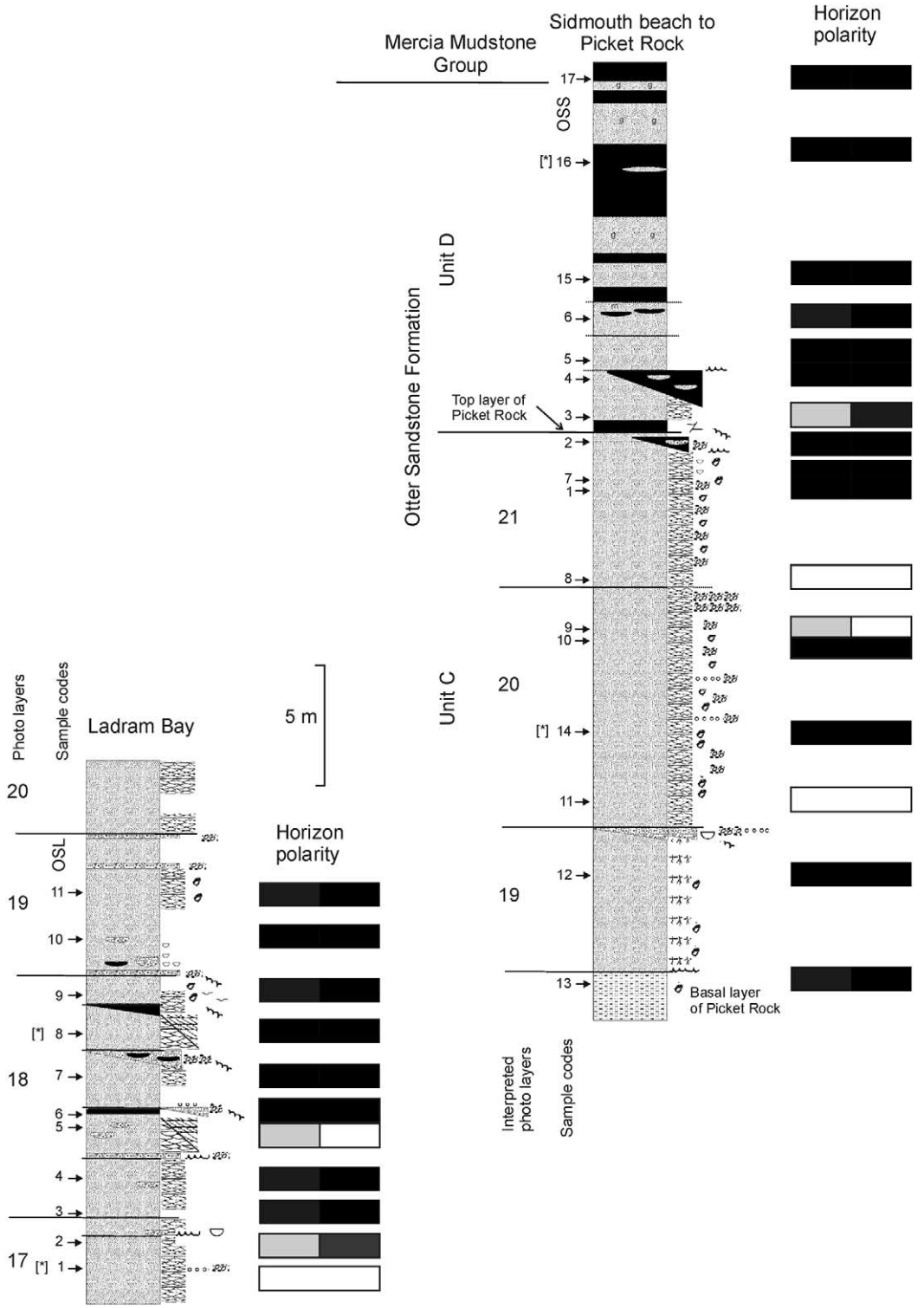


Fig. 5. Sedimentological log and sampling locations in the lowest part of the Otter Sandstone Formation east of the River Otter (lower part of unit B). The photo-layers are visually interpreted stratigraphic packages based on photographs of the cliff outcrop (see text for details). See Fig. 4 for key.

Fig. 6. Sedimentological log and sampling locations for the Otter Sandstone Formation, Poolness to Chiselbury Beach sections (Fig. 1B). See Fig. 4 for key.


Mercia Mudstone Group
Sidmouth beach to Picket Rock
Horizon polarity
Otter Sandstone Formation
Unit D
Unit C
OSS
Top layer of Picket Rock
Basal layer of Picket Rock
Interpreted photo layers
Sample codes
Photo layers
Sample codes
Ladram Bay
5 m
Horizon polarity
OSL

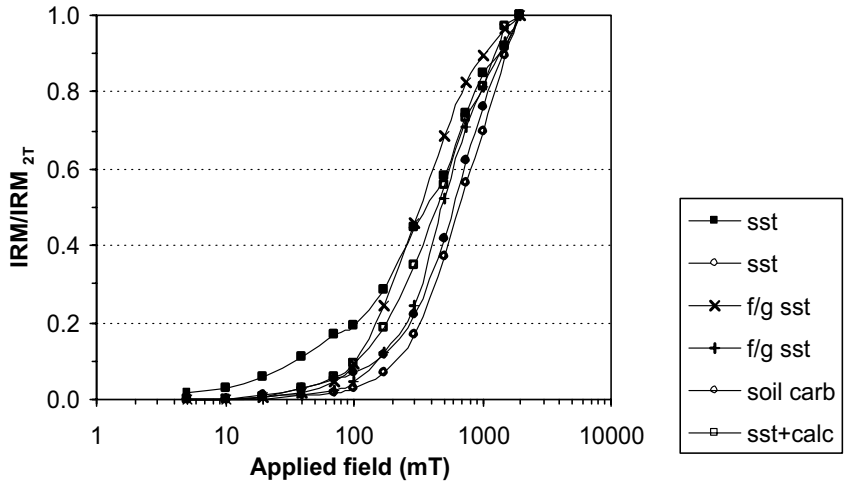


Fig. 8. IRM acquisition curves of representative lithologies. Abbreviations: sst, sandstone; f/g sst, fine-grained, muddy sandstone; soil carb, in situ rhizoconcretion; sst+calc, sandstone with calcrete.

caused by the heating. The specimen demagnetisation results were analysed using principle component analysis implemented in LINEFIND (Kent et al., 1983), which utilises the variance of the magnetisation measurement, available from the repeated magnetisation measurements on each specimen axis. The magnetic mineralogy of representative lithologies was studied using stepwise acquisition and thermal demagnetisation of Isothermal Remanent Magnetisation (IRM). IRM's were imparted using an ASC Scientific pulse magnetiser (model IM-30-10). Remanence measurements were made with a Molspin Ltd. spinner magnetometer, and thermal demagnetisation achieved using a Schonsted thermal demagnetiser. Susceptibility was measured after each step to monitor the thermally-induced mineralogical alteration.

4. Results

4.1. Magnetic properties

The IRM acquisition curves are all similar in shape (Fig. 8), acquiring the majority of IRM in fields greater than 100 mT, and not reaching saturation by 2 T. In some cases (e.g. upper sandstone sample in Fig. 8) an inflexion in the acquisition curve is seen at ~100 mT, which indicates the presence of a low coercivity fraction. On the basis of IRM results, orthogonal fields of 0.1 T and 2 T were used in the IRM demagnetisation experiments, with typical demagnetisation curves shown in Fig. 9. The 0.1 T fraction is always the smaller of the two components and generally exhibits maximum unblocking temperatures of 530–600°C, which is attributed to the presence of magnetite. The 0.1–2 T fraction exhibits a broad range of unblocking temperatures with maximum values of 650–710°C, which can be ascribed to haematite. In some instances there is a sharp drop in IRM between 30 and 100°C (e.g. Fig. 9B) that might be due to the presence of goethite. The lack of reddening of the internal parts of the calcrete root systems, the early diagenesis of these rootlet systems (Wright and Sandler, 1994) and the abundance of red mudstone fragments in the calcite cemented intraformational conglomerates,

Fig. 7. Sedimentological log and sampling locations in the uppermost Otter Sandstone Formation. See Fig. 4 for key. The location of the photo-layer boundaries in the Sidmouth Beach to Picket Rock (Fig. 1B) section are interpreted, since the photo-layers are only discernable as far NE as Ladram Bay.

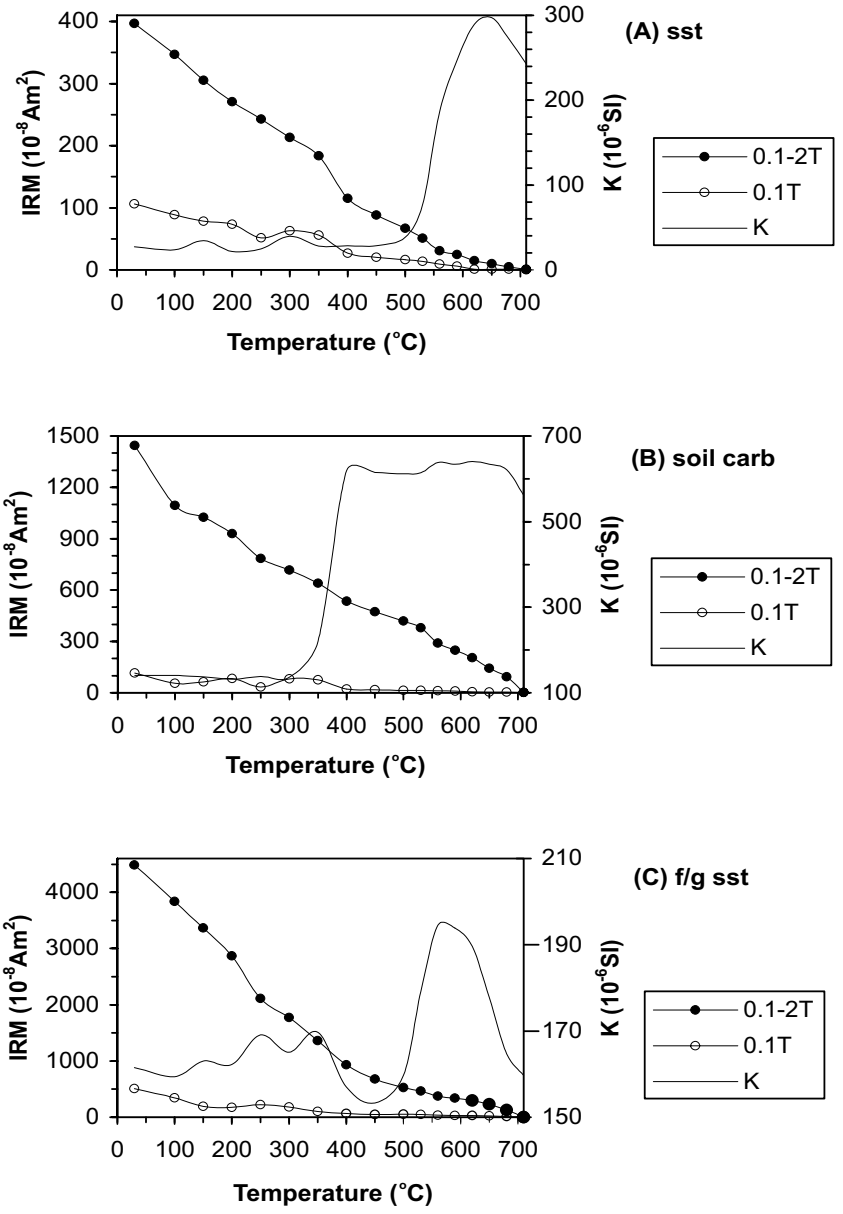


Fig. 9. Thermal demagnetisation of orthogonal IRMs. Lithologic abbreviations as in Fig. 8.

suggests that most of the haematite was produced early in the diagenetic history.

The specimens from unit D generally have greater NRM intensity and susceptibility than those from lower in the Otter Sandstone Formation (Fig. 10A,B), mainly because they are from mud-rich lithologies. The susceptibility of specimens from layers 16 and 17 in unit C (Fig. 6; 108–116 m in Fig. 10A), is somewhat greater than that of other samples from the Otter Sandstone Formation, even though the NRM intensity is not (Fig. 10A,B). This may be the result of a high paramagnetic content. Muddy and fine-grained sandstones tend to have higher suscepti-

bility and IRM (2 T) values, although there is no simple relationship between IRM and susceptibility. As with the NRM results, this may in part be explained by varying paramagnetic contributions to the susceptibility.

4.2. Specimen palaeomagnetic behaviour

From the 92 sampled horizons, 181 specimens (mostly two from each hand-sample) were demagnetised. The sandstone lithologies are generally less than ideal palaeomagnetic recorders. The specimens showed three magnetisation directions: a NE directed, intermediate to shallow characteristic magnetisation direction (ChRM) interpreted as Triassic normal polarity (Fig. 11A), a SW, mostly upwards, intermediate to shallow directed ChRM, interpreted as Triassic reverse polarity (Fig. 11B), and a steeper downwards directed magnetisation interpreted as a recent (Brunhes-age) magnetisation. The Triassic dual polarity magnetisations were isolated between 300 and 680°C, but most commonly above 500°C. At lower demagnetisation temperatures the recent magnetisation was often dominant. In all 5% of the samples were dominated by the Brunhes-age magnetisation up to either complete demagnetisation or instability caused by heating-induced mineralogy change. These overprinted specimens mostly occurred in the lower parts of the Sherwood Sandstone Group (outcrops west of River Otter). In some specimens the Brunhes-age overprint was present as both the low and a high temperature component ($> \sim 650$°C, but often ill-defined); in these specimens the Triassic magnetisation was isolated between 450 to 650°C (Fig. 11D).

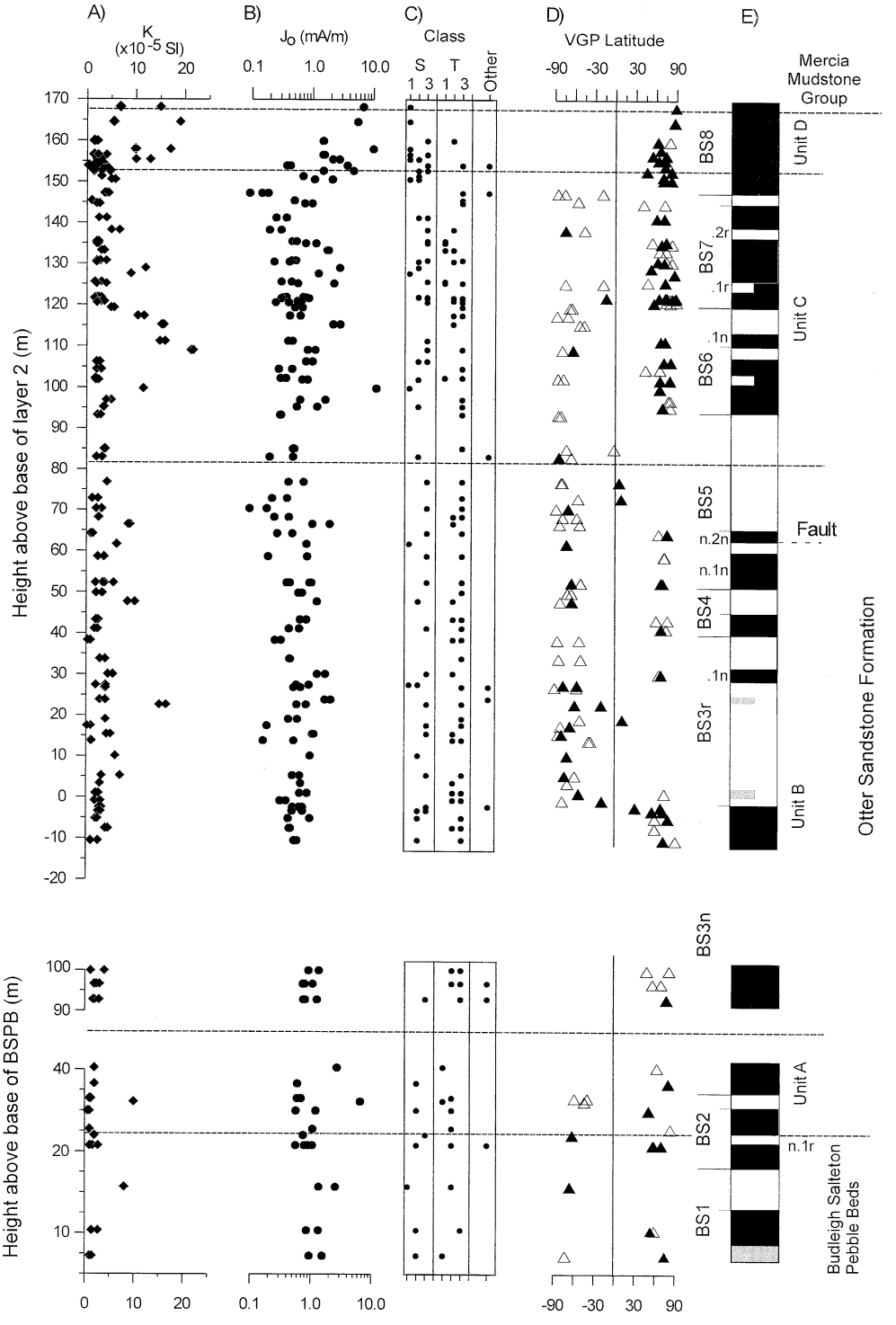
The behaviour of the specimens during demagnetisation was divided into two main classes, i.e. types S and T. S-type behaviour (38% of the specimens) indicates a linear trajectory which can be satisfactorily isolated from the demagnetisation vectors. These specimens were assigned a quality factor based on the noisiness of the principle component fits, with S1 (5% of all specimens; Fig. 11A,B) for best quality data, to S3 (17% of all specimens; Fig. 11C) for lowest quality data. In all 58% of the specimens showed motion along great-circle paths (T-type behaviour; Fig. 11D–F) towards either interpreted Triassic normal or reverse directions. The T-type behavioural specimens were also assigned a quality factor (T1, T2 or T3) based on a visual qualitative assessment of the noisiness and length of the great-circle paths (Fig. 10C). A small number of the specimens (4%) were directionally too unstable to be classified into these behaviour types.

As well as principal component line fits evaluated for all the S-type specimens (Fig. 11A–C), poles to the great-circle paths were evaluated from the T-type and some S-type specimens (e.g. Fig. 11D,E). A statistical test for planarity is used in LINEFIND to filter statistically imprecise great-circle planes (Kent et al., 1983). The majority of the fitted planes are origin intersection planes, although 40% did not include the origin, due mostly to mineralogical alteration and consequent viscous magnetisation acquisition at higher demagnetisation temperatures.

The Fisher mean directions of the S-type specimens have a mean at 021/33° ($\alpha_{95} = 5.1°$, $k = 12.4$, $n = 69$). Utilising both the fitted-lines from the S-type specimens, and the great-circle plane information from the T-type specimens in the McFadden and McElhinney (1988) combined great-circle analysis, gives a mean at 023/34° ($\alpha_{95} = 3.5°$, $k = 12.8$, $n = 138$). The means based on either the S-type data or the combined great-circle data pass the McFadden and McElhinney (1990) reversal test, with class b and c, respectively. A consistent low dip of 5–10° in all the sections precludes a fold test. The great-circle combined mean gives a virtual geomagnetic pole (VGP) at 52.8°N, 138.8°E for the Sherwood Sandstone Group. This is statistically the same as the Anisian VGP of Nawrocki and Szulc (2000) and the upper Anisian to lower Ladinian VGP of Theveniaut et al. (1992).

4.3. Magnetostratigraphy

For the S-type specimens the ChRM directions were converted into VGP latitude (Opdyke and Channell, 1996) with respect to the combined mean direction pole at 52.8°N, 138.8°E. This was also done for the T-type specimens, but using the point on each specimen's great-circle plane


A)
K (x10-5 SI)
0 10 20
B)
Jo (mA/m)
0.1 1.0 10.0
C)
Class
S T Other
1 3 1 3
D)
VGP Latitude
-90 -30 30 90
E)
Mercia Mudstone Group
Unit D
Unit C
Unit B
Unit A
Fault
Otter Sandstone Formation
Budleigh Salteton Pebble Beds
BS8
BS7
.2r
.1r
.1n
BS6
BS5
n.2n
n.1n
BS4
.1n
BS3r
BS3n
BS2
n.1r
BS1
Height above base of layer 2 (m)
Height above base of BSPB (m)

that was used in calculating the combined mean (Fig. 10A). This method of great-circle based VGP latitude calculation has been utilised also by Clement et al. (2000).

Based on its VGP latitude and the behavioural class, each specimen was assigned a polarity quality (R, R?, ?, N?, or N), to indicate its status. In some cases, a single specimen with good quality palaeomagnetic data (S1, S2, or T1) was sufficient to define the polarity of that horizon. However, since the majority of specimens are from sandstones, which are generally less than ideal palaeomagnetic recorders, more than one specimen was generally required from each horizon in order to confidently assign a horizon polarity (Figs. 4–7). Two horizons have specimens with opposite polarity, one in the Budleigh Salterton Pebble Beds (horizon B1; Fig. 4) and one in the Otterton Point section (horizon OSD11; Fig. 5).

The magnetozones resulting from the merging of the sub-section horizon polarities (Fig. 10E) are numbered (prefix BS to indicate the Budleigh Salterton to Seaton cliff sections) according to the protocol used for marine magnetic anomalies (Opdyke and Channell, 1996), except they proceed from older to younger (Figs. 10E and 12).

Within the Budleigh Salterton Pebble Beds only magnetozone BS2n.1n is defined by sampling at more than one horizon (Figs. 4 and 10), indicating the magnetostratigraphy through this unit is less than ideally defined. Unit A and the base of unit B of the Otter Sandstone Formation are dominated by normal polarity except for BS2r which is defined by sampling at two horizons (Fig. 10E). Two of the four normal magnetozones in the predominantly reverse polarity upper parts of unit B (i.e. BS3r.1n and BS5n.2n) are defined by only a single horizon (Fig. 10). The higher normal magnetozone, BS5n.2n, occurs adjacent to an array of faults with small displacements at the southern end of Poolness Beach. The photolayers cannot be traced with confidence through this fault array, and it is probable that BS5n.2n is a repetition of the upper part of BS5n.1n and is displayed as such in Fig. 13. Lithological correlation is of little help because of rapid lateral facies changes.

The upper part of unit C and all of unit D are dominated by normal polarity (BS6n to BS8n), with possibly six reverse polarity magnetozones in unit C, two of which (BS6n.1r and BS7n.1r) are defined by only a single specimen (half-width in Figs. 10E and 12). The stratigraphic overlap between the Ladram Bay and Picket Rock sections (Fig. 7) is compatible with the normal polarity of the overlapped intervals. Similarly, the stratigraphic overlap of the Chiselbury Beach section with the base of the Ladram Bay section is also consistent with the magnetostratigraphy (Figs. 5–7). The normal polarity present in unit D continues 1 m into the Mercia Mudstone Group, after which it is succeeded by a 30-m-thick reverse polarity magnetozone (Hounslow et al., unpublished records). Creer (1959) also sampled what is here termed unit D and the basal ca. 40 m of the overlying Mercia Mudstone Group and obtained a magnetostratigraphy consistent with our results.

5. Discussion

The absence of soil profiles, the presence of truncated phreatophytic rootlet systems and the relative absence of overbank fines suggest that a time interval is absent in the Otter Sandstone Formation stratigraphic record which is most clearly symbolised by the re-deposited calcretes in the channel-lag conglomerates. Red mudstone clasts in the channel lag conglomerates suggest the reddening and production of the haematite, comparable in time duration to the potential for preser-

Fig. 10. Summary magnetostratigraphic plot. (A) The magnetic susceptibility (k). (B) NRM_0 intensity (J_0). (C) The demagnetisation class of each specimen is represented by a dot, with the S-class representing planar fits, and T-class great-circle trends. Key: 1, best quality; 3, worst. (D) The VGP latitudes derived from the line fit ChRM (filled symbols) or points on the great-circle plane (unfilled symbols) for each specimen used in calculating the mean; values approaching +90° represent normal and those approaching −90° represent reversed polarity. (E) The composite magnetostratigraphy and magnetozone labels. The results from all measured specimens are recorded here.

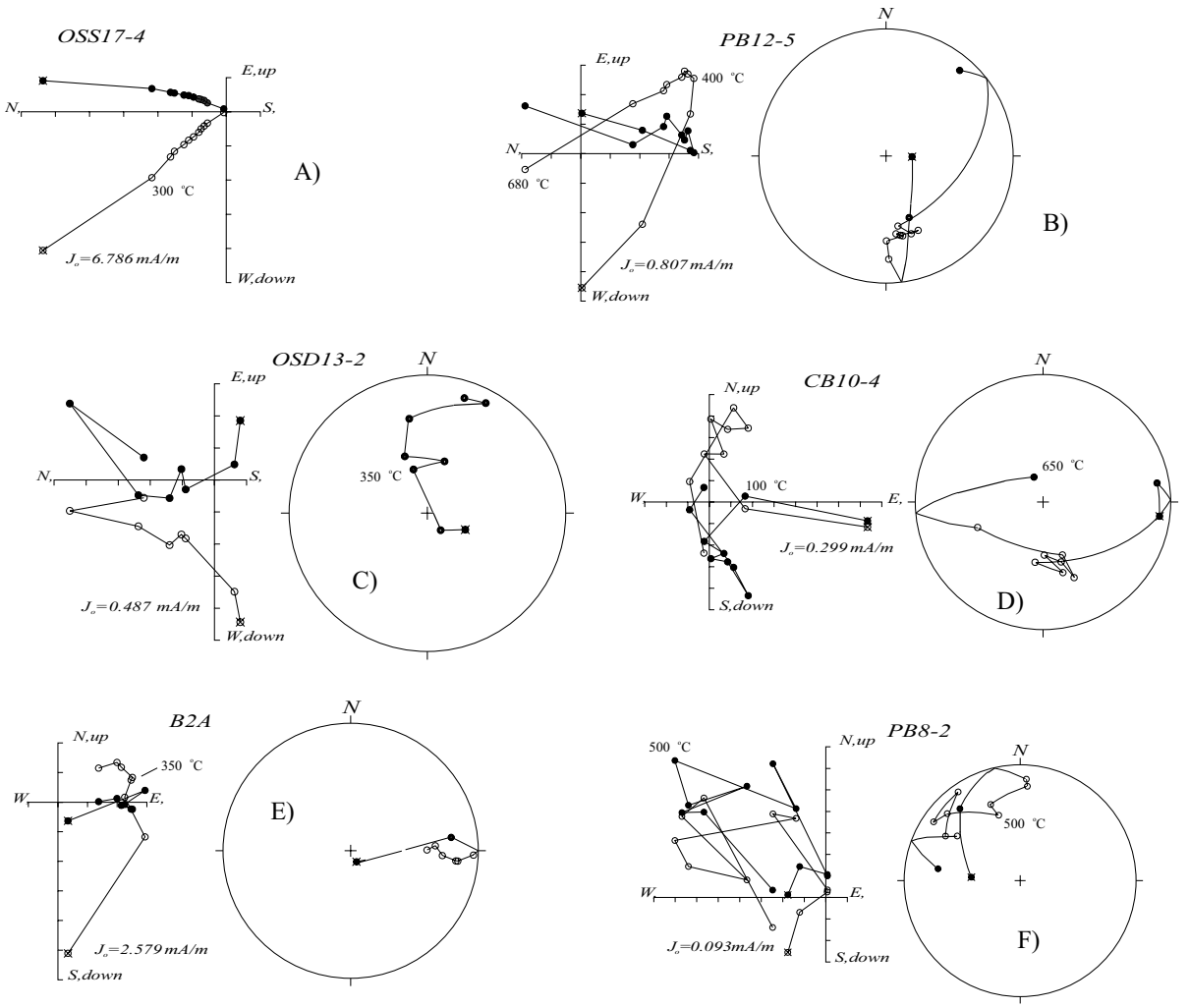


Fig. 11. Representative demagnetisation data from specimens. (A) Triassic normal polarity ChRM line fit (023°,+32°; S1 class) from 300°C through the origin (base of Mercia Mudstone Group). (B) Reverse polarity line fit (185°, −28°; S1 class) from 400–680°C (unit B, photo-layer 10). Both the low temperature and high temperature (ill defined >680°C) measurements appear to indicate a NNE oriented, downwards directed magnetisation, interpreted as possibly present day. (C) A poorly defined (chemical alteration sets in at 450°C) ChRM line fit (005°,+41°; class S3) from 350°C through the origin (unit B, layer 1). (D) Specimen CB10-4 (unit C, layer 15) has a fitted line between 540–650°C (157°, −51°; class T1), but also has a strong great-circle trend with a pole to the great-circle plane at 015°, −34° (100–650°C), indicating the line fit direction is probably composite. (E) Specimen B2A (Budleigh Salterton Pebble Beds) shows motion along a great-circle path (pole to great-circle plane at 329°, −7°; class T2) due to incomplete separation of a steep downwards-directed magnetisation (present day component) and a component with probable negative inclination. (F) Specimen PB8-2 (unit B, layer 11) shows erratic motion along a great-circle path towards a westerly direction. The great-circle path is best defined (from 500°C through origin; class T3) with a great-circle pole at 063°, −53°.

vation of a channel sandstone package. Tauxe et al. (1980) have demonstrated this in other fluvial red beds. The effects of the duration of diagenetic and channel erosional events on the fidelity of the magnetostratigraphy, depend partly upon the sedimentation rate. If sediment accumulated relatively rapidly in the channel-related packages (bounded by planar surfaces 1–4 m apart), in

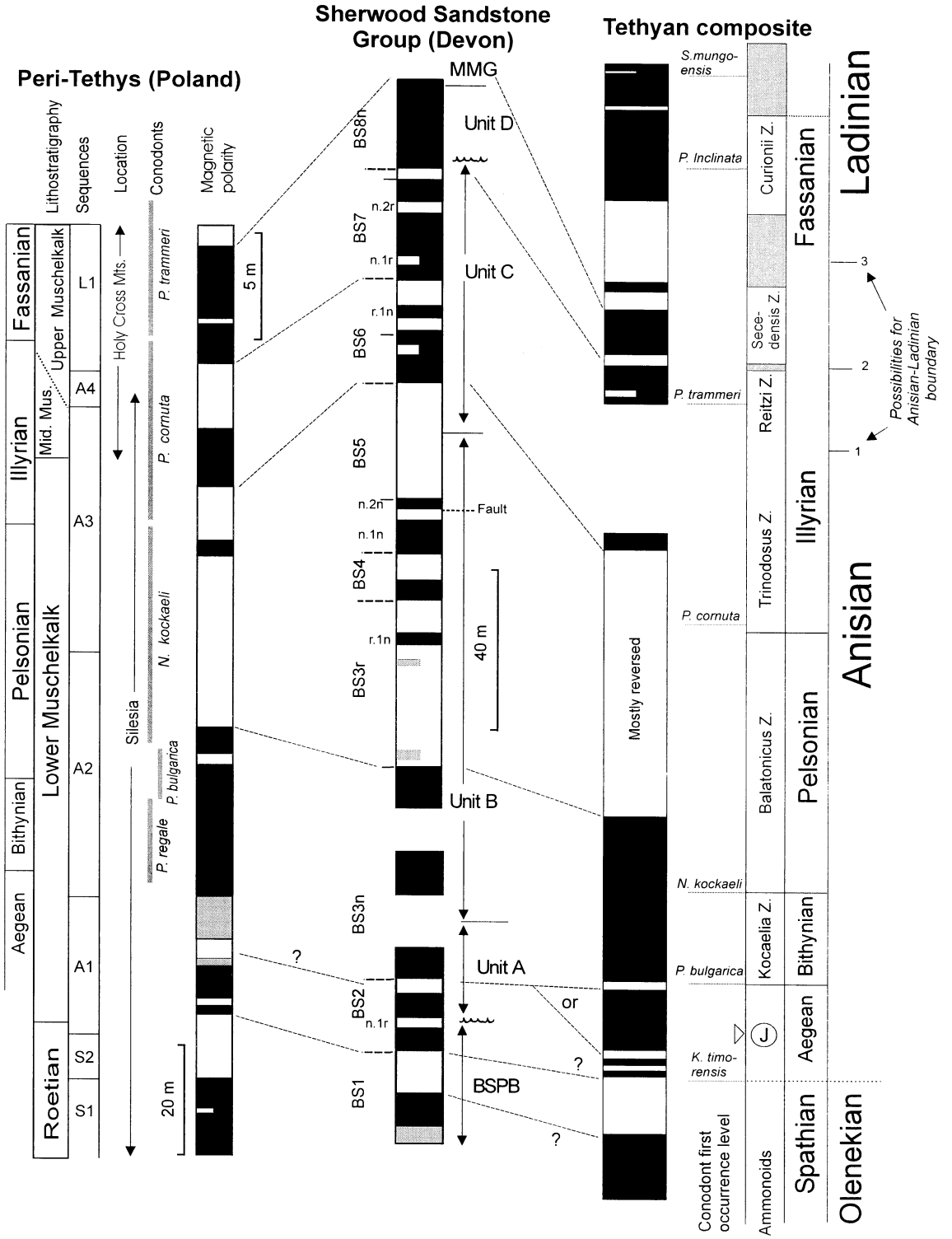


Fig. 12 (*Caption overleaf*).

comparison to the polarity changes, the overall style of those changes will be preserved. Perhaps not all short polarity periods may be preserved, and others may be truncated, but the change between mixed polarity periods to longer duration single polarity intervals will be apparent. The means of testing such truncation could be to look in detail at the location of the polarity boundaries in relation to the bounding surfaces of the channel-related packages in those parts of the Otter Sandstone Formation where polarity changes are common.

More extended hiatuses in fluvial systems represent more of a problem since these may be substantial time gaps. The lower boundary of the Otter Sandstone Formation may represent such a hiatus. Seismic and well studies in the Wessex Basin suggest that the major disconformities are at the base of the Budleigh Salterton Pebble Beds and the base of the Otter Sandstone Formation (Butler, 1998). Within the deeper parts of the Wessex Basin an additional unit (not seen at outcrop) appears to be present between the reflector marking the base of the Otter Sandstone Formation and the equivalent of the Budleigh Salterton Pebble Beds. There is no evidence for extended hiatuses within the Otter Sandstone Formation.

The Anisian age for the Otter Sandstone Formation fauna is based partly on its similarity to a fauna from the Upper Buntsandstein (Röt Formation) from which Anisian conodonts have been recovered (Benton, 1997; Lucas, 1998). However, the tetrapod fauna does share elements that are potentially upper Olenekian to Ladinian in age (Benton, 1997). Lucas (1998) goes further, suggesting that his Perovkan land vertebrate faunachron is not likely to be younger than lower Anisian. Therefore, on the basis of the tetrapod data the Otter Sandstone Formation may represent the whole of the Anisian, with perhaps some overlap into the Olenekian and Ladinian, or it may be older than Middle Anisian.

5.1. *Magnetostratigraphic comparison to the marine record*

The distinctive feature of the Otter Sandstone Formation magnetostratigraphy is the predominantly normal BS2n–BS3n magnetozone interval, followed above by the dominantly reverse BS3r to BS5r magnetozones. This interval would appear to match closely (Fig. 12) the Aegean to lower Illyrian interval found in both the Tethyan composite compiled by Muttoni et al. (2000) and the Polish Muschelkalk (Nawrocki and Szulc, 2000). With this as the basis of a correlation, the upper part of unit C and all of unit D can best be matched to the upper Illyrian to lower Fassanian interval from the Polish Muschelkalk (Fig. 12).

Muttoni et al. (2000) identified one normal polarity interval and two single-sample magnetozones in the lower Illyrian of the Edivetur section, which may be the equivalent of BS4n and BS5n. An additional normal magnetozone in the upper Pelsonian of the Edivetur section may be the equivalent of BS3r.1n. The Tethyan composite magnetostratigraphy of Muttoni et al. (2000) has a data gap over the middle–upper Illyrian interval which appears to correspond to most of magnetozone BS6. Huang and Opdyke (2000) have shown that an Illyrian major normal magnetozone (this appears to correspond to BS6n) follows a middle Pelsonian to lower Illyrian predominantly reverse magnetozone. This normal magnetozone appears to be the equivalent of the middle Illyrian normal

Fig. 12. Sherwood Sandstone Group magnetostratigraphy compared to biostratigraphically calibrated Tethyan composite magnetostratigraphy (Muttoni et al., 2000) and the Polish Muschelkalk (Nawrocki and Szulc, 2000). Thickness scales (Holy Cross Mountains and Silesia sections) and sequence stratigraphic units shown in Peri-Tethys (Polish) data. The Tethyan composite magnetostratigraphy is compiled from several sections and has no single thickness scale. The Tethyan ammonite biostratigraphy uses the biostratigraphy published with the sections (see Muttoni et al., 2000) and consequently uses different zonal schemes for the upper and lower parts (Brack and Reiber, 1993; Mietto and Manfrin, 1995). Some conodont genus names have been changed conforming to Muttoni et al. (2000). The base of the Aegean has been placed according to the conodont biostratigraphy from the Kçira section (Muttoni et al., 1997b). J indicates the level of the Aegeiceras–Japonites beds in the Chios section projected onto the Kçira section magnetostratigraphy (Muttoni et al., 1997b).

magnetozone in the Polish Muschelkalk, and BS6n (Fig. 12).

In marine Tethyan sections the upper Illyrian to lower Fassanian interval is characterised by the dominance of normal polarity (Muttoni and Kent, 1994; Muttoni et al., 1997a; Marton et al., 1997; Gallet et al., 1998). The magnetostratigraphy from the overlying 140 m of the Mercia Mudstone Group (Hounslow et al., unpublished records) shows a good match to the composite Ladinian polarity record of Muttoni et al. (2000), indicating that the top of unit D (just below the BS8n–BS8r transition; Fig. 13) is correlated to the middle of the Secedensis Zone of Brack and Reiber (1993), identified in the Frötschbach section (Muttoni et al., 1997a). The reinterpretation by Brack et al. (2001) of the Margon section magnetostratigraphy allows the BS8n–BS8r polarity transition to be correlated with the Sepianensis Subzone (Nevadites Zone) of Mietto and Manfrin (1995).

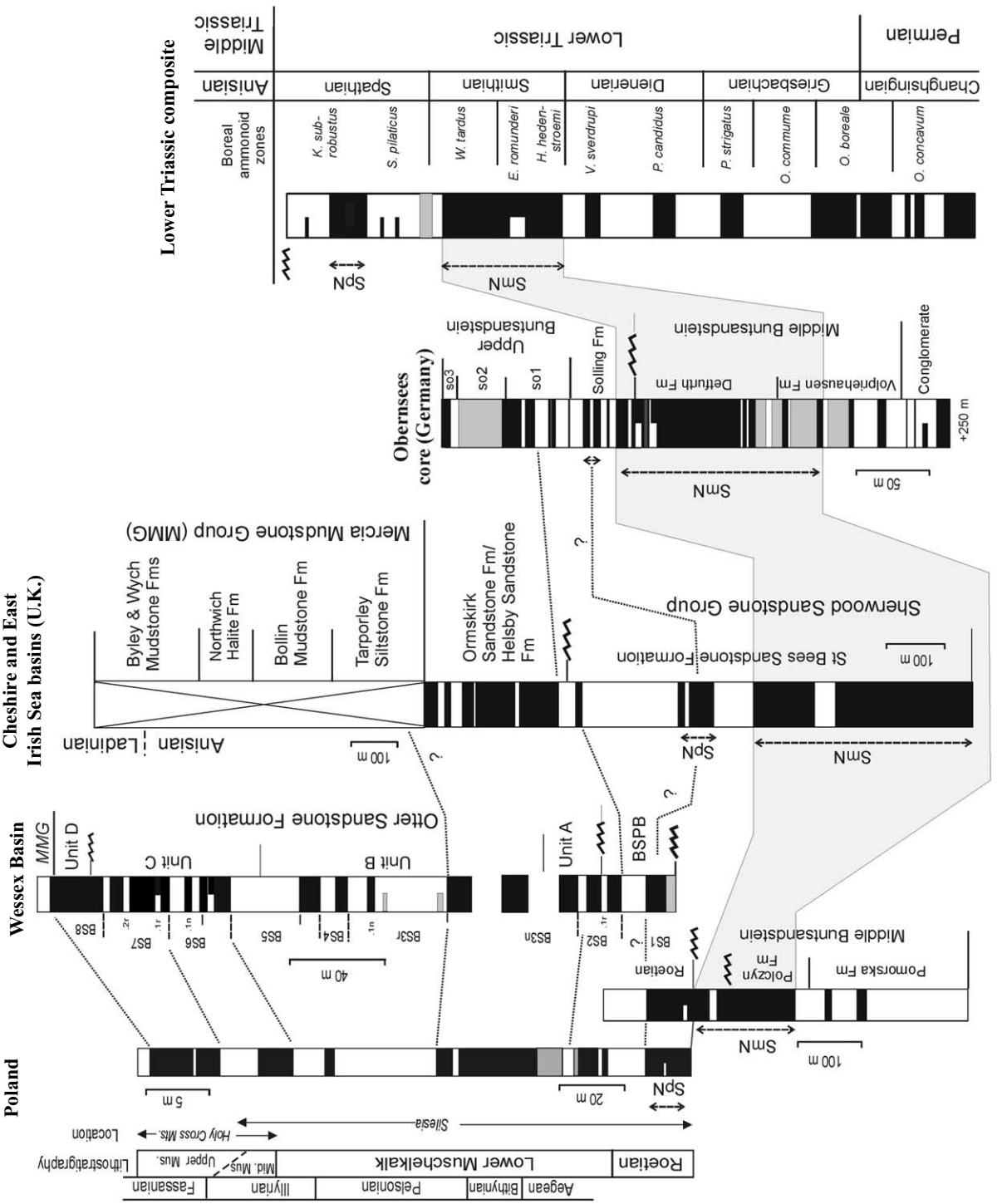
Within the upper part of unit C (BS7; Fig. 12) there are rather more short duration reverse polarity intervals in comparison to the record from the Polish Muschelkalk, which may be a reflection of the rather condensed Upper Muschelkalk sections measured by Nawrocki and Szulc (2000). These magnetostratigraphic correlations of the Otter Sandstone Formation indicate that the Perovkan land vertebrate faunachron is equivalent to most of the Anisian, rather than just the lower Anisian.

5.2. Regional relationships

In the Cheshire Basin the topmost unit of the Sherwood Sandstone Group, the Helsby Sandstone Formation (Figs. 1A and 13), has Anisian tetrapods and palynomorphs (Benton and Spencer, 1995; Warrington et al., 1999). Within the eastern Irish Sea Basin (Fig. 1A), the Helsby Sandstone Formation equivalent, the Ormskirk Sandstone Formation, is dominated by normal polarity (Mange et al., 1999). Palynology indicates the Anisian–Ladinian boundary to be located within the Mercia Mudstone Group, near the boundary between the Byley and Wych Mudstone formations (Fig. 13; Warrington et al., 1999). This suggests that the normal polarity in the Ormskirk Sandstone Formation is the approximate equivalent of BS3n in the Sherwood Sandstone Group of the Devon coast, and, consequently, that the base of the Otter Sandstone Formation and Ormskirk Sandstone Formation are age equivalent. Data from undemagnetised specimens, measured by Clegg et al. (1954), show mixed polarity for the lower part of the Mercia Mudstone Group in the Cheshire and Carlisle Basins, consistent with the correlation in Fig. 13.

Magnetostratigraphic correlations of the Lower Triassic within Europe are impeded by the paucity of constraining biostratigraphy and the presence of several unconformities, such as the Hardegsen disconformity near the top of the Middle Buntsandstein (Aigner and Bachman, 1992; Nawrocki, 1997; Geluk and Rohling, 1999). Persistent features of Lower Triassic magnetostratigraphy are the Smithian and middle Spathian normal polarity magnetozones (Ogg and Steiner, 1991; Lozovsky and Molostovsky, 1993; Steiner et al., 1993; Hounslow et al., 1996; Muttoni et al., 1997b). The Smithian normal magnetozone (SmN) probably occurs in the upper part of the Middle Buntsandstein (Nawrocki, 1997; Soffel and Wippern, 1998), and the lower part of the St Bees Sandstone Formation (Fig. 13; Mange et al., 1999; Rey, 1992). The middle Spathian normal magnetozone (SpN) appears to be present in the Roetian of Poland, the upper part of the St Bees Sandstone Formation and in the Obernsees core (Fig. 13).

The unconformity between the Ormskirk Sandstone and St Bees Sandstone formations (and equivalents in the Cheshire Basin; Evans et al., 1993; Jackson et al., 1995) has been correlated with the Hardegsen disconformity of Germany and the southern North Sea Basin (Warrington et al., 1980; Cameron et al., 1992; Evans et al., 1993). Within the eastern Irish Sea Basin this unconformity is barely perceptible (Meadows and Beach, 1993; Jackson et al., 1995), suggesting that the St Bees Sandstone and Ormskirk Sandstone formations may contain a reasonably complete representation of the Smithian to lower Anisian magnetostratigraphy. This would seem to be


Lower Triassic composite
Middle Triassic
Lower Triassic
Permian
Anisian
Spathian
Smithian
Dienerian
Griesbachian
Changhsingian
Boreal ammonoid zones
K. sub-robustus
S. pilaticus
W. tardus
E. romunderi
H. heden-stroemi
V. sverdrupi
P. candidus
P. strigatus
O. commune
O. boreale
O. concavum
SpN
SmN
Obernsees core (Germany)
Upper Buntsandstein
Middle Buntsandstein
Solling Fm
Detfurth Fm
Volpriehausen Fm
Conglomerate
so3
so2
so1
+250 m
50 m
Cheshire and East Irish Sea basins (U.K.)
Mercia Mudstone Group (MMG)
Byley & Wych Mudstone Fms
Northwich Halite Fm
Bollin Mudstone Fm
Tarporley Siltstone Fm
Ormskirk Sandstone Fm/ Helsby Sandstone Fm
Sherwood Sandstone Group
St Bees Sandstone Formation
Anisian
Ladinian
100 m
Wessex Basin
MMG
Unit D
Unit C
Unit B
Unit A
BSPB
Otter Sandstone Formation
BS8
BS7
BS6
BS5
BS4
BS3r
BS3n
BS2
BS1
40 m
Poland
Roetian
Middle Buntsandstein
Polczyn Fm
Pomorska Fm
5 m
20 m
Silesia
Holy Cross Mts.
Location
Lithostratigraphy
Lower Muschelkalk
Mid. Mus.
Upper Mus.
Aegean
Bithynian
Pelsonian
Illyrian
Fassanian

born out by the magnetostratigraphic correlations (Fig. 13), indicating the disconformities at the base of the Ormskirk Sandstone Formation and the Otter Sandstone Formation may be approximately equivalent with an interval in the Röt Formation of the Obernsees core (approximately sequence 5 of Aigner and Bachman, 1992). These correlations would seem to refute the idea that the Hardegsen disconformity is the equivalent of the basal Otter Sandstone Formation (and Ormskirk Sandstone Formation) disconformity.

The magnetostratigraphy alone cannot constrain the age of the Budleigh Salterton Pebble Beds, since it is bounded by disconformities. The sedimentology and sheet-like nature of the Budleigh Salterton Pebble Beds in the Wessex Basin, suggest it represents the deposits of a river system in a slowly subsiding basin, with a distant source and deposition not significantly affected by local basin tectonics (Smith, 1990; Smith and Edwards, 1991). Therefore, the Budleigh Salterton Pebble Beds could represent either a significant part of the Lower Triassic, or the initial phase of subsidence immediately prior to deposition of the much thicker Otter Sandstone Formation.

The Conglomérate Principal (20–30 m thick) in the Paris Basin (Goggin and Jacquin, 1998) is a unit similar to the Budleigh Salterton Pebble Beds; it occurs below palaeosols, is a coarse-grained fluvial conglomerate, and is derived from the Armorican Massif in western France. The far-sourced nature of the clastics of the Budleigh Salterton Pebble Beds and the basin evolution arguments presented by Smith and Edwards (1991) suggest that both these conglomerates are the response to events in the Variscan uplands in Brittany and the southern English Channel. The Conglomérate Principal caps the Paris Basin equivalent of the Buntsandstein. Consequently, the Budleigh Salterton Pebble Beds are most likely to be equivalent to an interval near the top of the Buntsandstein. The good magnetostratigraphic correlation to the Roetian of the Polish Basin (Fig. 13) suggests the Budleigh Salterton Pebble Beds are probably equivalent to the middle Spathian to lower Aegean (Figs. 12 and 13), perhaps almost continuous with deposition of the Otter Sandstone Formation. Wright et al. (1991) estimate a minimum of 50 000–100 000 years to form the reg palaeosol at the upper boundary of the Budleigh Salterton Pebble Beds, a time duration which does not violate the correlations proposed.

North American strata possessing tetrapods of the Perovkan land vertebrate faunachron are the Holbrook and Anton Chico members of the Moenkopi Formation in Arizona and New Mexico (Lucas, 1998). Both members display relatively simple polarity patterns, with a normal at the base followed above by mostly reverse, with a single shorter normal (Molina-Garza et al., 1991, 1996). In comparison to the magnetostratigraphy of the Otter Sandstone Formation these members may represent only a part of the Anisian, perhaps the Aegean to lower Illyrian.

6. Conclusions

The palaeomagnetic signal from the Sherwood Sandstone Group in Devon preserves a dual polarity magnetisation, which is mostly carried by haematite. The magnetisations pass the reversal test and give a mean palaeopole direction, which is the same as other Middle Triassic palaeomagnetic data from cratonic Europe. A comparison to the magnetostratigraphy from marine sections, with better biostratigraphy indicates that the Otter Sandstone Formation preserves a mostly complete record of the Anisian magnetic field polarity. This indicates the Perovkan land vertebrate faunachron of Lucas (1998) is equivalent to most of the Anisian.

The age of the Budleigh Salterton Pebble Beds

Fig. 13. Comparison of the Sherwood Sandstone Group magnetostratigraphy between the Devon coast, the Cheshire and eastern Irish Sea basins (Mange et al., 1999), the Polish Buntsandstein and Muschelkalk (Nawrocki, 1997; Nawrocki and Szulc, 2000) and the Buntsandstein in the Obernsees core (Soffel and Wippern, 1998). The Upper Permian and Lower Triassic timescale is from Hounslow et al. (1996), Gallet et al. (2000) and Yin et al. (2001). Abbreviations: SmN, Smithian Normal; SpN, Spathian Normal.

is not tightly constrained, but its similar sedimentology to the Conglomérate Principal of the Paris Basin, combined with the magnetostratigraphy, indicates it may have been deposited during the middle Spathian (upper Olenekian) to lower Aegean (lower Anisian). The disconformity that marks the base of the Otter Sandstone Formation is equivalent to the boundary between the St Bees Sandstone and Ormskirk Sandstone formations in the eastern Irish Sea Basin, but these are not equivalent with the Germanic Basin Hardegsen disconformity. Magnetostratigraphic comparison to the eastern Irish Sea Basin confirms that the top of the Sherwood Sandstone Group is diachronous, being within the lower Anisian in the eastern Irish Sea Basin. The magnetostratigraphy indicates the base of the Otter Sandstone Formation is within the Aegean (lowest stage of the Anisian) and the boundary between the Otter Sandstone Formation and the overlying Mercia Mudstone Group is located within the Nevadites Zone of Mietto and Manfrin (1995).

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